

Signal-to-Interference-Noise Ratio (SINR) and Signal-to-Noise Ratio (SNR) Improvement in the Massive Multiple-Input Multiple-Output (MIMO) Systems

Garima Kulshreshtha¹ and Dr. Usha Chauhan²

Department of Electronics and Communication Engineering
School of Electrical, Electronics and Communication Engineering, Galgotias University,
Greater Noida (Gautam Buddh Nagar), India

¹garima.kulshreshtha.gk@gmail.com, ²usha.chauhan@galgotiasuniversity.edu.in

Abstract—Millimeter-wave wireless communications concepts have been considered as a key technology for future 5G communication systems. Analog or digital hybrid precoders and combiners architectures have been used to reduce the propagation loss in the MIMO systems. The combiner or receive combiner bank (RCB) of the massive MIMO systems used to regenerate the transmitted symbols or signals through the different types of optimizations techniques and algorithms. This paper presents the updated model of the massive MIMO receiver system to enhance the SINR and SNR of the traditional massive MIMO systems. The updated receiver system has least-mean-square (LMS) adaptive filters to cancel the additive white Gaussian noise (AWGN) to improve the SINR and SNR of the massive MIMO system.

Keywords—Massive MIMO; RCB; SINR; SNR; AWGN; LMS Adaptive Filter; Noise Cancellation;

I. INTRODUCTION

Massive MIMO based communication systems are common in trends for 5G wireless systems [1], [2]. The massive MIMO technology has the several benefits as very large antenna array [3], high throughput [4], decrease computational complexity [5], better SINR [6], mitigate inter-cell interference [5], [7], spectral efficiency [8], [9], ultra-low latency [10], high-speed field [11], special division multiple access [12], etc. This technology is focused on a large number of users presented in the small region because the communication system has a higher channel capacity [13]. The spatial multiplexing gives a higher spectral efficiency due to a large number of the transmitting and receiving antennas [14]. Precoder and combiner are the most important parts of the massive MIMO communication systems. Hybrid form is the advanced form of precoder and combiner [8], [12], [15]–[17]. The hybrid precoder and combiner design is another key area in the massive MIMO systems. The hybrid precoder and combiner have merits like- effective channel gain [18], low computational complexity, higher sum-rate [19], maximize SNR, maximize SINR [20], inter primary base station interference [1], etc. This paper includes the mathematical analysis of the proposed massive MIMO system. The MIMO systems have several methods [21], [22] from the last decade

to enhance the performance of RCB [6], [23]. The enhancing parameters for the RCB of the massive MIMO systems are SINR, SNR, bit-error-rate, packet-error-rate, etc [24].

In traditional massive MIMO receiver system, RCB receives the input signal with channel noise, resulting in low SINR so this paper investigates a new model of the massive MIMO receiver system to reduce or eliminate the massive MIMO channel noise using adaptive filters which adapts the behaviour of channel and eliminates the channel noise from the output of the linear transformation (LT) receiver. The proposed model also enhances the SINR and SNR of the massive MIMO receiver systems.

II. MASSIVE MIMO SYSTEM MODEL

The massive MIMO systems [21], [24], [25] have the N_t number of transmitting antennas and the N_r number of receiving antennas shown in figure 1. The system is presenting the output signal vector [22] as per the given relation in the equation (1).

$$y = Hs + n \quad (1)$$

Where, s is the complex input signal vector with dimension $N_t \times 1$, H is the massive MIMO channel matrix with dimension $N_r \times N_t$, n is the additive white Gaussian noise (AWGN) vector of the MIMO channel with dimension $N_r \times 1$ and y is the output signal vector with dimension $N_r \times 1$. The signals s , n , y and H are shown in the equations (2), (3), (4) and (5) respectively.

$$s = [s_1 \ s_2 \ s_3 \ \dots \ s_p \ \dots \ s_{N_t}]^T \quad (2)$$

Where, s_p is the p^{th} transmitted signal or symbol and $1 \leq p \leq N_t$. $(.)^T$ denotes the transpose of the given vector.

$$n = [n_1 \ n_2 \ n_3 \ \dots \ n_q \ \dots \ n_{N_r}]^T \quad (3)$$

Where n_q is the received channel noise signal component at the q^{th} receiving antenna and $1 \leq q \leq N_r$.

$$y = [y_1 \ y_2 \ y_3 \ \dots \ y_q \ \dots \ y_{N_r}]^T \quad (4)$$

Where, y_q is the q^{th} output signal and $1 \leq q \leq N_r$.

$$H = \begin{bmatrix} H_{1,1} & H_{1,2} & H_{1,3} & \dots & H_{1,p} & \dots & H_{1,N_t} \\ H_{2,1} & H_{2,2} & H_{2,3}^c & \dots & H_{2,p} & \dots & H_{2,N_t} \\ H_{3,1} & H_{3,2} & H_{3,3}^c & \dots & H_{3,p} & \dots & H_{3,N_t} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ H_{q,1} & H_{q,2} & H_{q,3} & \dots & H_{q,p} & \dots & H_{q,N_t} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ H_{N_r,1} & H_{N_r,2} & H_{N_r,3} & \dots & H_{N_r,p} & \dots & H_{N_r,N_t} \end{bmatrix} \quad (5)$$

Where, $H_{q,p}$ is the channel matrix coefficient between the q^{th} receiving antenna and the p^{th} transmitting antenna for $1 \leq p \leq N_t$ and $1 \leq q \leq N_r$.

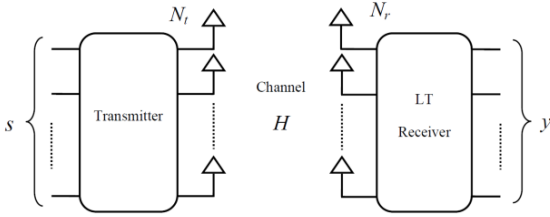


Fig. 1. Block diagram of Massive MIMO communication system

The AWGN must have zero mean value and covariance value σ_n^2 . The input signal symbol also has a covariance value σ_s^2 . The receiver must have complete knowledge of the channel. The channel matrix coefficients are independent and identically distributed (i.i.d.) complex Gaussian distribution function [2], [26]. The block diagram of the RCB of the massive MIMO receiver system is shown in figure 2. The output signal generated with the help of LT receiver pass into the RCB and the RCB regenerate the transmitted input symbol vector which is equivalent to the output signal vector of the RCB. The output signal vector α of the RCB is given in equation (6). Equation (6) is also presented in the form of channel matrix H , input symbol, or signal vector s and channel noise n using equation (1).

$$\alpha = C^T y = C^T H s + C^T n \quad (6)$$

Where C is the receive combining matrix (RCM) with dimension $N_r \times N_r$. The RCM is described in equation (7).

$$C = \begin{bmatrix} C_{1,1} & C_{1,2} & C_{1,3} & \dots & C_{1,p} & \dots & C_{1,N_t} \\ C_{2,1} & C_{2,2} & C_{2,3}^c & \dots & C_{2,p} & \dots & C_{2,N_t} \\ C_{3,1} & C_{3,2} & C_{3,3}^c & \dots & C_{3,p} & \dots & C_{3,N_t} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ C_{q,1} & C_{q,2} & C_{q,3} & \dots & C_{q,p} & \dots & C_{q,N_t} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ C_{N_r,1} & C_{N_r,2} & C_{N_r,3} & \dots & C_{N_r,p} & \dots & C_{N_r,N_t} \end{bmatrix} \quad (7)$$

Where, $C_{q,p}$ is the channel matrix coefficient between the q^{th} receiving antenna and the p^{th} transmitting antenna for $1 \leq p \leq N_t$ and $1 \leq q \leq N_r$.

The p^{th} vector component of the output of RCB of the massive MIMO system [21] is described in equation (8).

$$\alpha_p = C_p^T H_p s_p + C_p^T \sum_{u=1, u \neq p}^{N_t} H_u s_u + C_p^T n \quad (8)$$

Where, RCM C and channel matrix H are presented in the column matrix form as shown in equations (9) and (10) respectively.

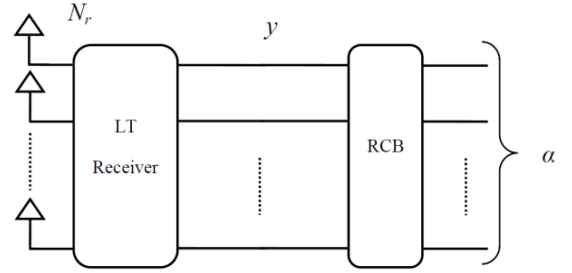


Fig. 2. Block diagram of Massive MIMO receiver system with RCB

$$C = [C_1 \ C_2 \ C_3 \ \dots \ C_p \ \dots \ C_{N_t}] \quad (9)$$

The C_p is the p^{th} column vector component of the RCM C with dimension $N_r \times 1$.

$$H = [H_1 \ H_2 \ H_3 \ \dots \ H_p \ \dots \ H_{N_t}] \quad (10)$$

The H_p is the p^{th} column vector component of the channel matrix H with dimension $N_r \times 1$. The p^{th} received signal component SINR ($SINR_{p,trad}$) of the traditional massive MIMO receiver system is also described in the equation (11).

$$SINR_{p,trad} = \frac{E \left[\|C_p^T H_p s_p\|^2 \right]}{E \left[\|C_p^T \sum_{u=1, u \neq p}^{N_t} H_u s_u\|^2 \right] + E \left[\|C_p^T n\|^2 \right]} \quad (11)$$

The p^{th} received signal component SNR ($SNR_{p,trad}$) of the traditional massive MIMO receiver system is also described in the equation (12).

$$SNR_{p,trad} = \frac{E\left[\left\|C_p^T H_p s_p\right\|^2\right]}{E\left[\left\|C_p^T n\right\|^2\right]} \quad (12)$$

The SINR and SNR of the traditional massive MIMO receiver system depend on channel AWGN. The proposed work of the model of this paper is based on the elimination of the channel AWGN. This proposed model will enhance the SINR and SNR of the traditional massive MIMO receiver system.

III. PROPOSED WORK

The proposed work of this paper is to cancel the massive MIMO channel noise using a combination of LMS adaptive filters [27], [28] configured between LT receiver and RCB unit of the massive MIMO receiver system. The selection of LMS filter is as per their simple algorithm which has low computational complexity. The process of noise cancellation using the LMS adaptive filter is known as an adaptive noise cancellation (ANC) [29]–[31]. The reference input noise is compulsory to cancel the channel noise from the output of the LT receiver, so we are using the channel noise detection to detect the channel noise only at the receiver end. The proposed massive MIMO system has added a combination of adaptive filters between the LT receiver and RCB to cancel the channel noise with the help of the reference noise signal. The reference noise signal is transferred to an adaptive filter with the help of a channel noise detector and receiving antennas. This will improve the performance of the massive MIMO receiver system. The proposed MIMO model has a combination of N_r adaptive filters. The improved parameter can be accessed in the form of SINR after the RCB unit.

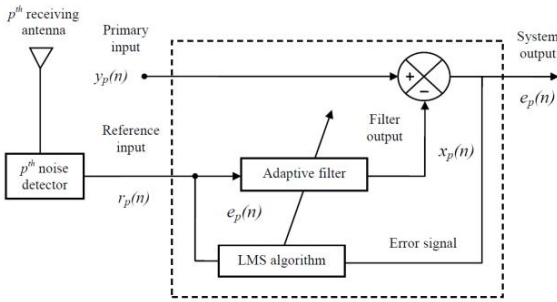


Fig. 3. Block diagram of the ANC system for the p^{th} component of the output signal of the p^{th} adaptive filter

In this section, we are reducing or approximately eliminating the p^{th} component of noise $n_p(n)$ with the help of the ANC and p^{th} component of reference channel noise signal $r_p(n)$ collected with the help of channel noise detectors. The proposed model has an N_r number of noise detectors to collect the reference noise signals from the N_r number of receiving antennas. The block diagram model of the ANC is shown in

figure 3 [32], [33]. The finite-impulse response (FIR) based adaptive filter adopts the behaviour of the noise with the help of reference input signal [34] or reference noise signal $r_p(n)$ and cancel the noise from the primary input signal $y_p(n)$ which is noisy information signal [29]. The reference noise signal $r_p(n)$ is generated with the help of p^{th} detector which is connected from the p^{th} receiving antenna. The adaptive filter output signal $x_p(n)$ is shown in equation (13).

$$x_p(n) = W_p^T(n) R_p(n) \quad (13)$$

Where, $W_p^T(n)$ is the filter coefficient vector and $R_p(n)$ is the reference input signal vector.

The signals vectors $W_p^T(n)$ and $R_p(n)$ are given in the equations (14) and (15) respectively.

$$W_p(n) = [w_{p,0}(n) \ w_{p,1}(n) \ \dots \ w_{p,L-1}(n)]^T \quad (14)$$

$$R_p(n) = [r_p(n) \ r_p(n-1) \ \dots \ r_p(n-L+1)]^T \quad (15)$$

Where, L is the filter length.

The p^{th} output signal $y_p(n)$ will be the same as given in equation (16) in the discrete signal form.

$$y_p(n) = \sum_{i=1}^{N_t} H_{p,i}(n) s_i(n) + n_p(n) \quad (16)$$

The p^{th} component of the system output of the ANC is shown in equation (17).

$$e_p(n) = y_p(n) - x_p(n) \quad (17)$$

Weights of the adaptive filter adjusted with the help of an adaptive filter algorithm to subtract the noise from the primary input signal as per the algorithm given in equation (18). The primary input signal is also known as the desired signal. The minimization of noise is based on the minimum mean square error (MMSE) techniques.

$$W_p(n+1) = W_p(n) + \mu R_p(n) e_p(n) \quad (18)$$

Where, μ is the step-size and $W_p(n+1)$ is the updated weight vector of the LMS adaptive filter.

IV. RESULT AND DISCUSSION

The proposed work cancels the p^{th} component of channel AWGN from the p^{th} LT receiver output of the massive MIMO communication system using the MMSE technique of LMS adaptive filter [35], [36]. The p^{th} component of the output of the ANC system is shown in equation (19).

$$e_p(n) = \sum_{i=1}^{N_t} H_{p,i}(n) s_i(n) + n_p(n) - x_p(n) \quad (19)$$

Let,

$$\sum_{i=1}^{N_t} H_{p,i}(n)s_i(n) = h_p(n) \quad (20)$$

Therefore, using equations (19) and (20),

$$e_p(n) = h_p(n) + n_p(n) - x_p(n) \quad (21)$$

The MSE is presented in equation (22).

$$E[e_p^2(n)] = E[h_p^2(n)] + E[n_p(n) - x_p(n)]^2 + 2E[h_p(n)\{n_p(n) - x_p(n)\}] \quad (22)$$

The noise signals $n_p(n)$ and $x_p(n)$ are correlated because both are the same channel noise, measured by the different hardware/system but signal $h_p(n)$ and noise signals are uncorrelated. Therefore, equation (22) will be simplified and given in equation (23).

$$E[e_p^2(n)] = E[h_p^2(n)] + E[n_p(n) - x_p(n)]^2 \quad (23)$$

The expectation of the uncorrelated signals is zero. By adjusting the adaptive filter's weights using LMS adaptive filter algorithms, we can minimize the MSE and the MMSE is shown in equation (24).

$$E[e_p^2(n)]_{\min} = E[h_p^2(n)] \quad (24)$$

Therefore, the condition for the MMSE is $n_p(n) = x_p(n)$. So, equation (24) concludes and presents the ANC system output as shown in equation (25).

$$e_p(n) \approx h_p(n) = \sum_{i=1}^{N_t} H_{p,i}(n)s_i(n) \quad (25)$$

The equation (25) gives the p^{th} LMS adaptive filter's system output which has no channel noise. This channel noise has been received by the p^{th} receiving antenna of the proposed massive MIMO receiver system. We use the N_r number of LMS adaptive filters to cancel all the noise factors, present in the received signal y at the LT receiver end which is shown in equation (1). The combination of the N_r number of LMS adaptive filters is used. The output vector e includes all the output components of all the LMS adaptive filters, given in equation (26).

$$e \approx Hs \quad (26)$$

In the traditional massive MIMO system, the input signal vector of RCB is signal vector y which is equal to $Hs+n$ and including the channel noise vector n also. In the proposed massive MIMO system, the input signal vector of RCB is signal vector e which is equal to Hs only and excluding the channel noise vector n . The output vector β of the RCB is given in equation (27). Equation (27) is also presented in the

form of channel matrix H and input symbol or signal vector s using equation (26).

$$\beta = C^T e \approx C^T Hs \quad (27)$$

The p^{th} vector component of the output of RCB of the proposed massive MIMO receiver system β_p is described in the equation (28).

$$\beta_p \approx C_p^T H_p s_p + C_p^T \sum_{u=1, u \neq p}^{N_t} H_u s_u \quad (28)$$

The p^{th} received signal component SINR ($SINR_{p,prop}$) of the proposed massive MIMO receiver system is also described in the equation (29).

$$SINR_{p,prop} \approx \frac{E[\|C_p^T H_p s_p\|^2]}{E[\|C_p^T \sum_{u=1, u \neq p}^{N_t} H_u s_u\|^2]} \quad (29)$$

The equations (11) and (29) present the SINRs of the traditional and proposed massive MIMO receiver system respectively and investigated that $SINR_{d,prop}$ is greater than $SINR_{d,trad}$ because the channel noise n is approximately eliminated from the overall output of the propose massive MIMO receiver system.

$$SINR_{p,prop} = \frac{E[\|C_p^T H_p s_p\|^2]}{E[\|C_p^T \eta\|^2]} \approx \infty \quad (30)$$

The equations (16) and (25) present the p^{th} inputs of RCB for traditional and proposed massive MIMO receiver systems. Therefore, the SNRs of the p^{th} inputs of RCB for traditional and proposed massive MIMO receiver systems are $SINR_{p,trad}$ and $SINR_{p,prop}$ respectively. These SNRs are given in the equations (12) and (30). These equations also present that the input signal of RCB for the proposed massive MIMO system has higher SNR concerning the traditional massive MIMO system. Ideally, the SNR of the proposed massive MIMO system will be infinite but practically it will be very high.

V. CONCLUSION

The massive MIMO communication systems have higher spectral efficiency as well as better SINR then conventional wireless communication systems. The proposed work is to introduce the LMS adaptive filter between the LT receiver block and the RCB unit as shown in figure 3. The LT receiver block is used to generate the output signal vector y which is given in the equation (1). This output signal vector y also has AWGN of the massive MIMO channel. The LMS adaptive filters eliminate the channel noise n and the output signal vector e of the LMS adaptive filters pass to the RCB. The

output signal vector of LMS adaptive filters eliminates the massive MIMO channel noise and passes to the RCB of the proposed massive MIMO system. If RCB has an input signals vector with no channel AWGN, it increases the SINR also. The equations (11) and (29) present the SINR of the p^{th} component of the output vector of the RCB of the traditional and proposed massive MIMO receiver system respectively. This paper concludes that the proposed system eliminates the channel AWGN and increases the SINR of the massive MIMO receiver system using a combination of LMS adaptive filters. This paper also concludes that the proposed massive MIMO system has a very high SNR concerning the traditional massive MIMO system as compared to equations (12) and (30). Therefore, the proposed massive MIMO receiver system will be more efficient concerning the traditional massive MIMO receiver system.

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